UNIVERSITÀ DI TRENTO

Performance Evaluation of TARP in Unstable Wireless Environments

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TARP: Tree-based Any-to-Any Routing Protocol

- Gradient-based routing paradigm
- Hybrid link cost: RSSI + ETX
  - RSSI: Received Signal Strength Indicator
  - ETX: Expected Transmission Count
- Reactive parent switching

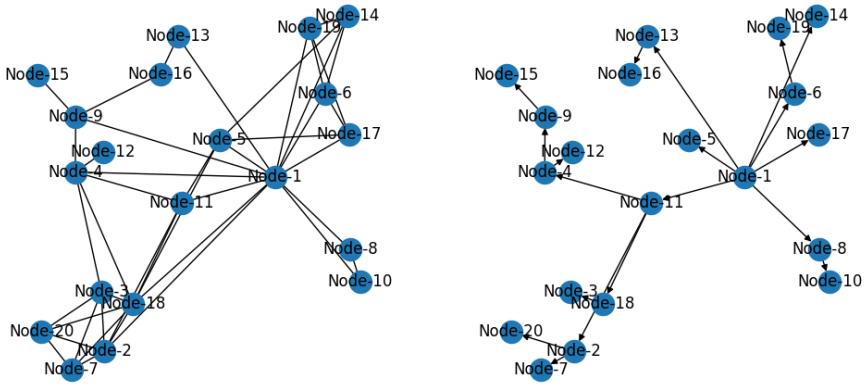


Figure: Connectivity graph (left) and routing tree (right) in random topology under lossy channel conditions.

Motivation

- TARP was designed for WSNs in a static environment
- We want to evaluate its performance under realistic conditions:
  - Different physical topologies (linear, grid, star, ring, random)
  - Different channel conditions
- With a tilt towards unfavorable conditions

Simulation Architecture: Custom DES

- Python-based discrete event simulator
- Single-threaded scheduler (parallelized runs)
- Modular architecture:
  - Node module (static, dynamic)
  - PHY module (channel modeling)
  - MAC module (IEEE 802.15.4)
  - Routing module (TARP implementation)
  - Attachable monitors for debugging and data collection
  - RNG module
- High-fidelity PHY + IEEE 802.15.4 MAC

Physical Layer Modeling

- Log-distance path loss
- Correlated shadowing (Gudmundson model)
- Nakagami-m fading
- SINR-based packet decoding

Here it would possibly be nice to have a visualization of something. The correlated shadowing feels like a good candidate.

MAC and Interference Modeling

- IEEE 802.15.4 Unslotted CSMA/CA
- Backoff + Retransmissions
- Capture Effect
- CCA via energy detection

Experimental Design

- 5 Topologies

  [Linear, Grid, Star, Ring, Random]
- 6 Channel Profiles

  [Ideal, Stable, Stable (Mid PL), Stable (High PL), Lossy, Unstable]
- Poisson Traffic Model ($\frac{1}{\lambda} = 60s$)
- 50 ARV pairs (100 runs per configuration)

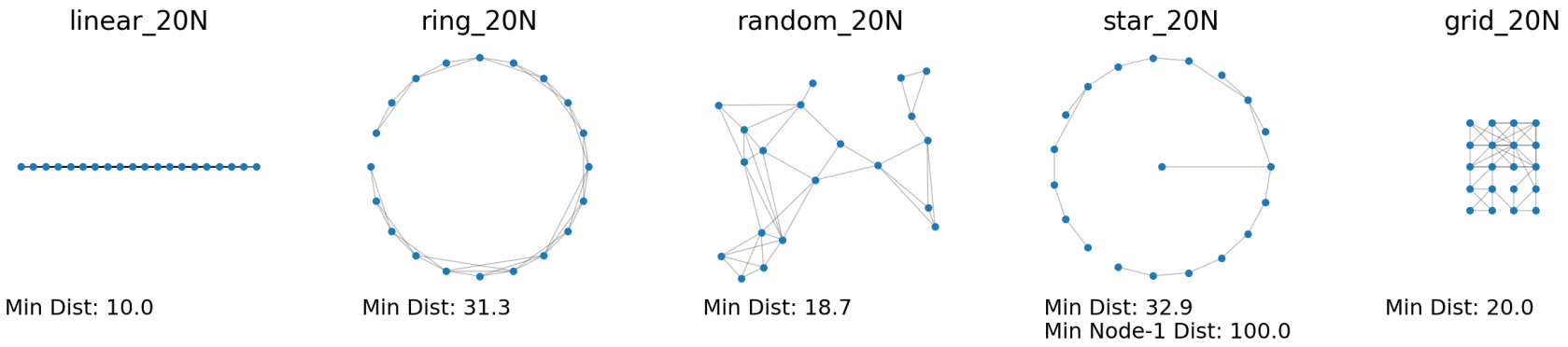


Figure: The 5 topologies considered in the evaluation

Statistical Methodology

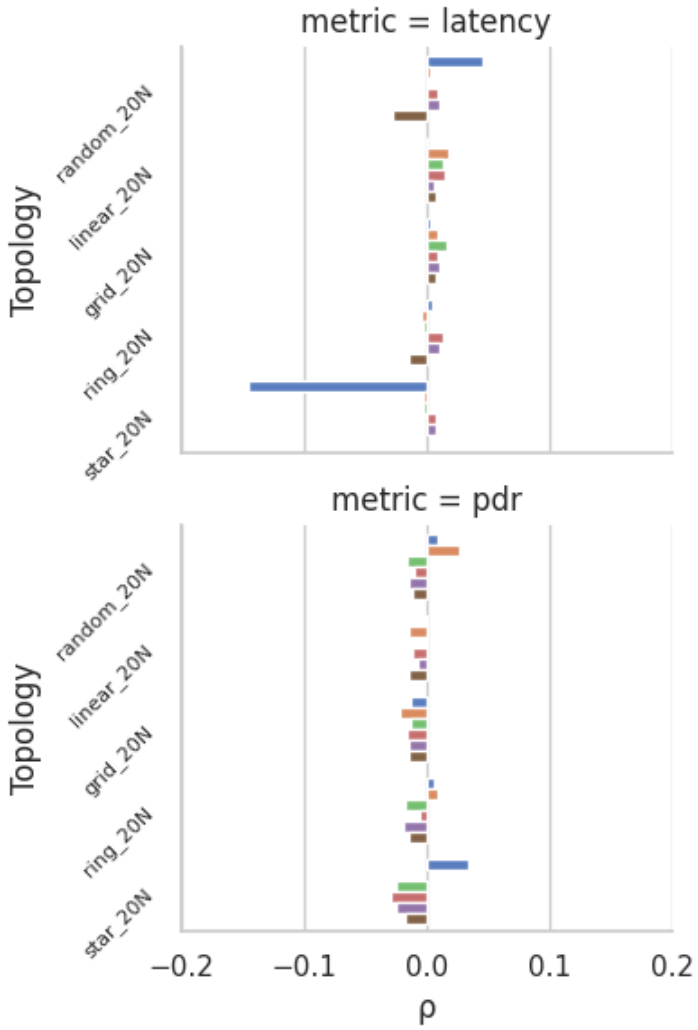
- Independent random streams (Philox PRNG)

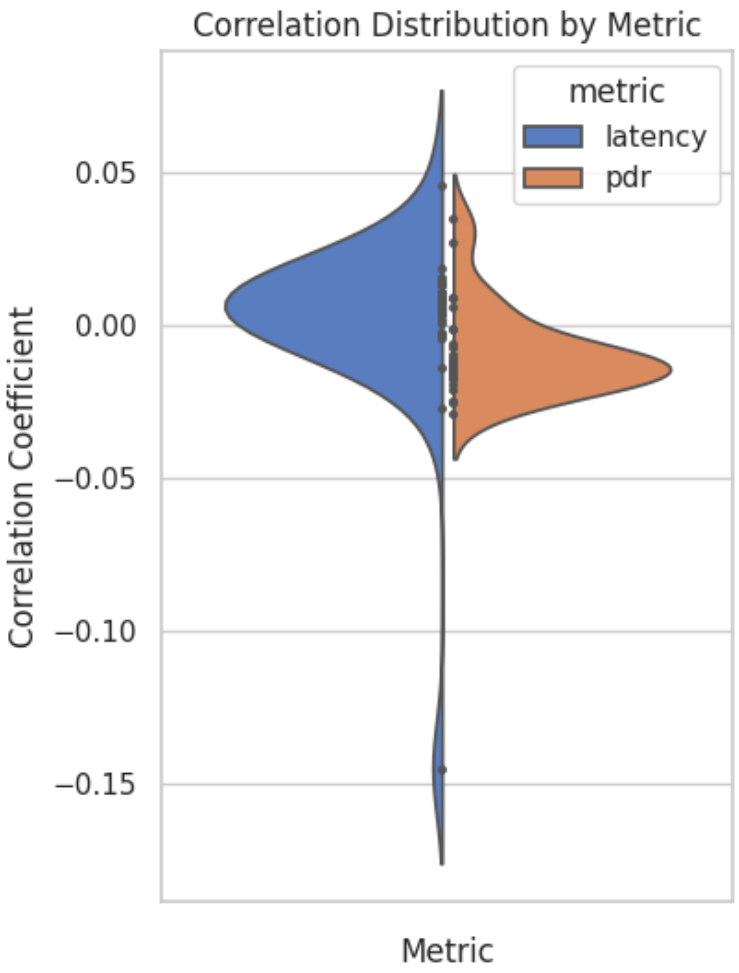
```
random_manager.get_stream(key = "NARROWBAND_CHANNEL/SHADOWING")
```

- Antithetic Random Variates (ARV)
  - $F^{-1}(U)$ and $F^{-1}(1-U)$
  - Uniform, Normal, Exponential and Nakagami-m have a closed-form inverse CDF
- Common Random Numbers (CRN) across configurations
- Bootstrap 95% Confidence Intervals

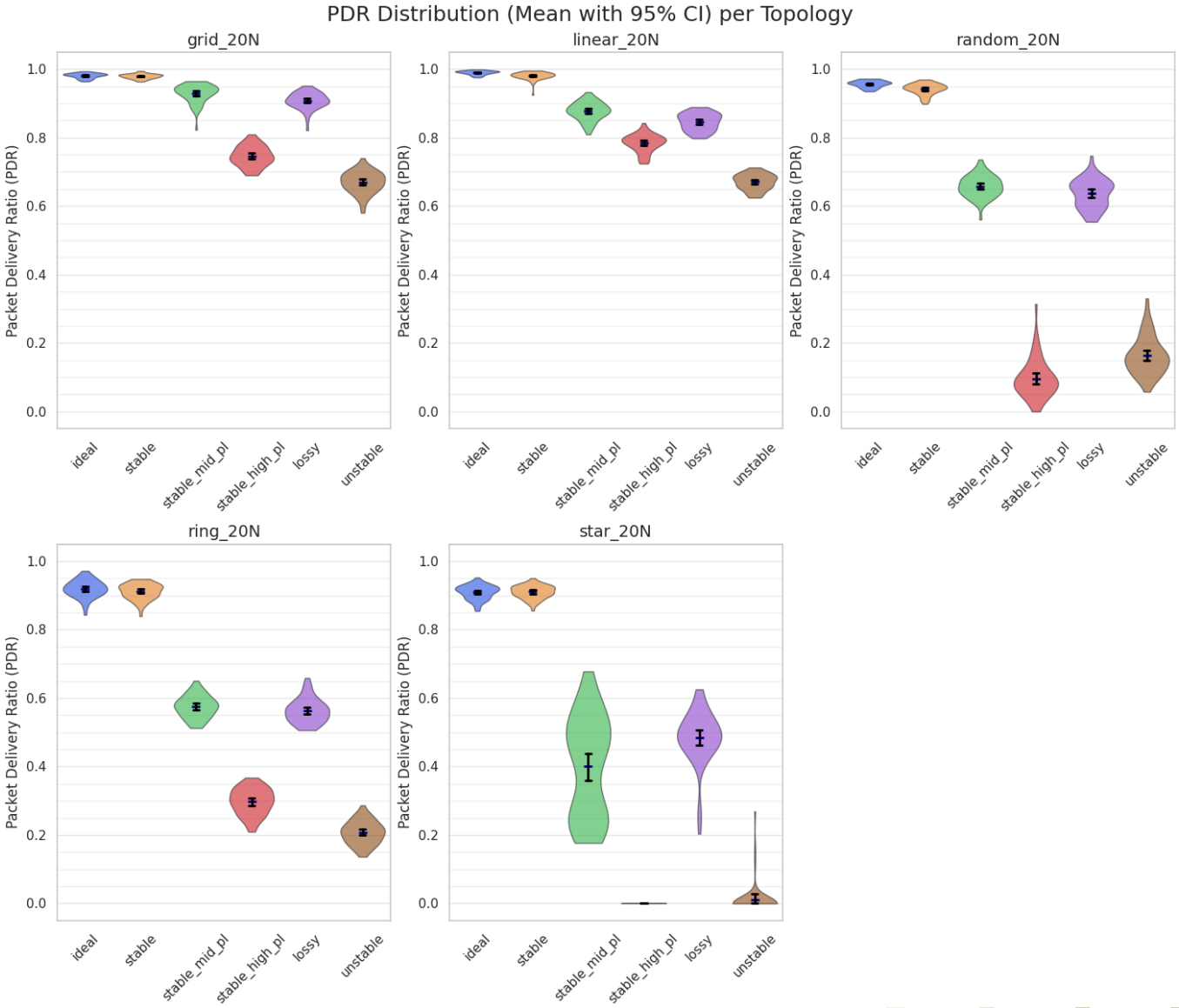
Control Variates

Interarrival time as a control variate.

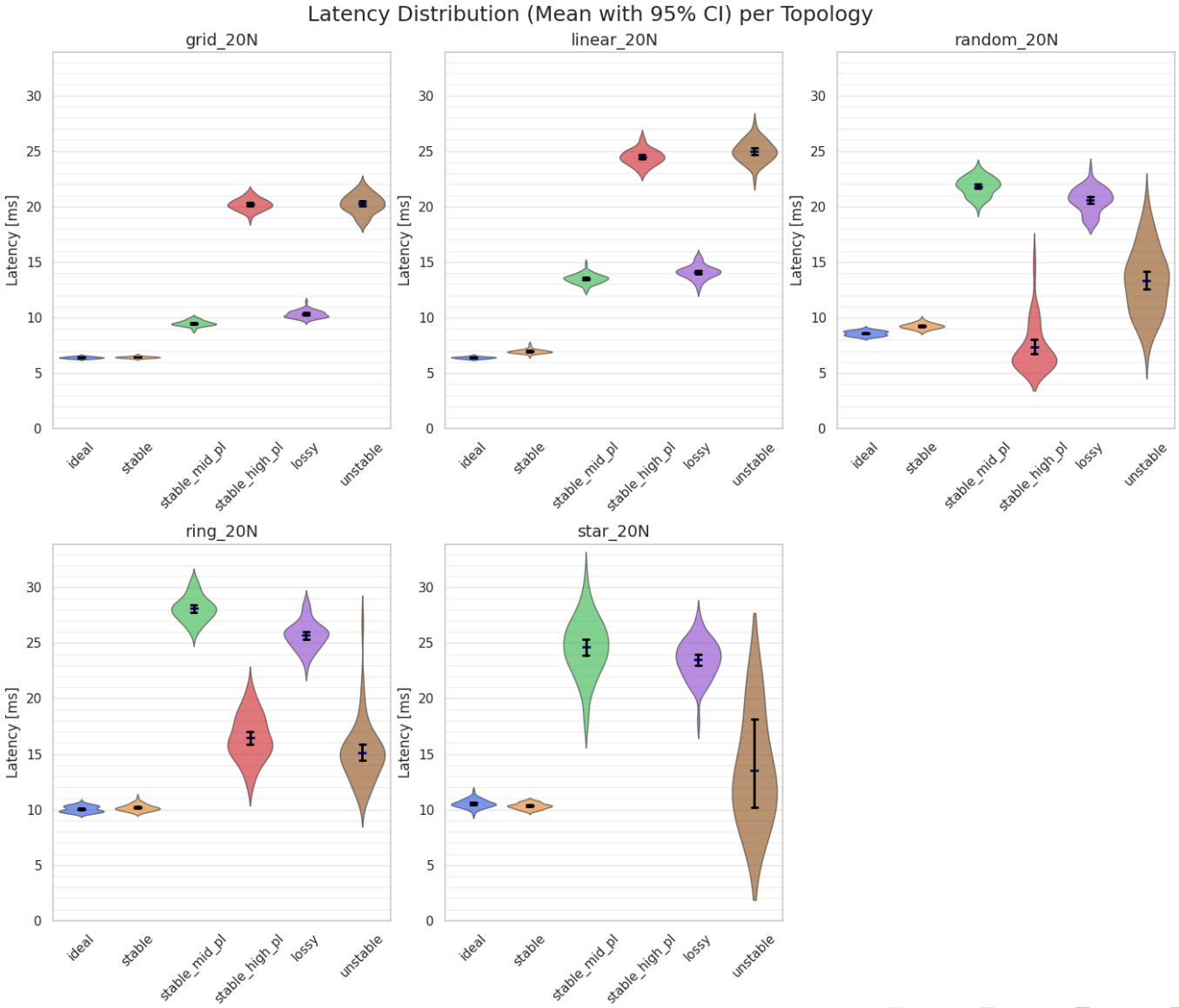




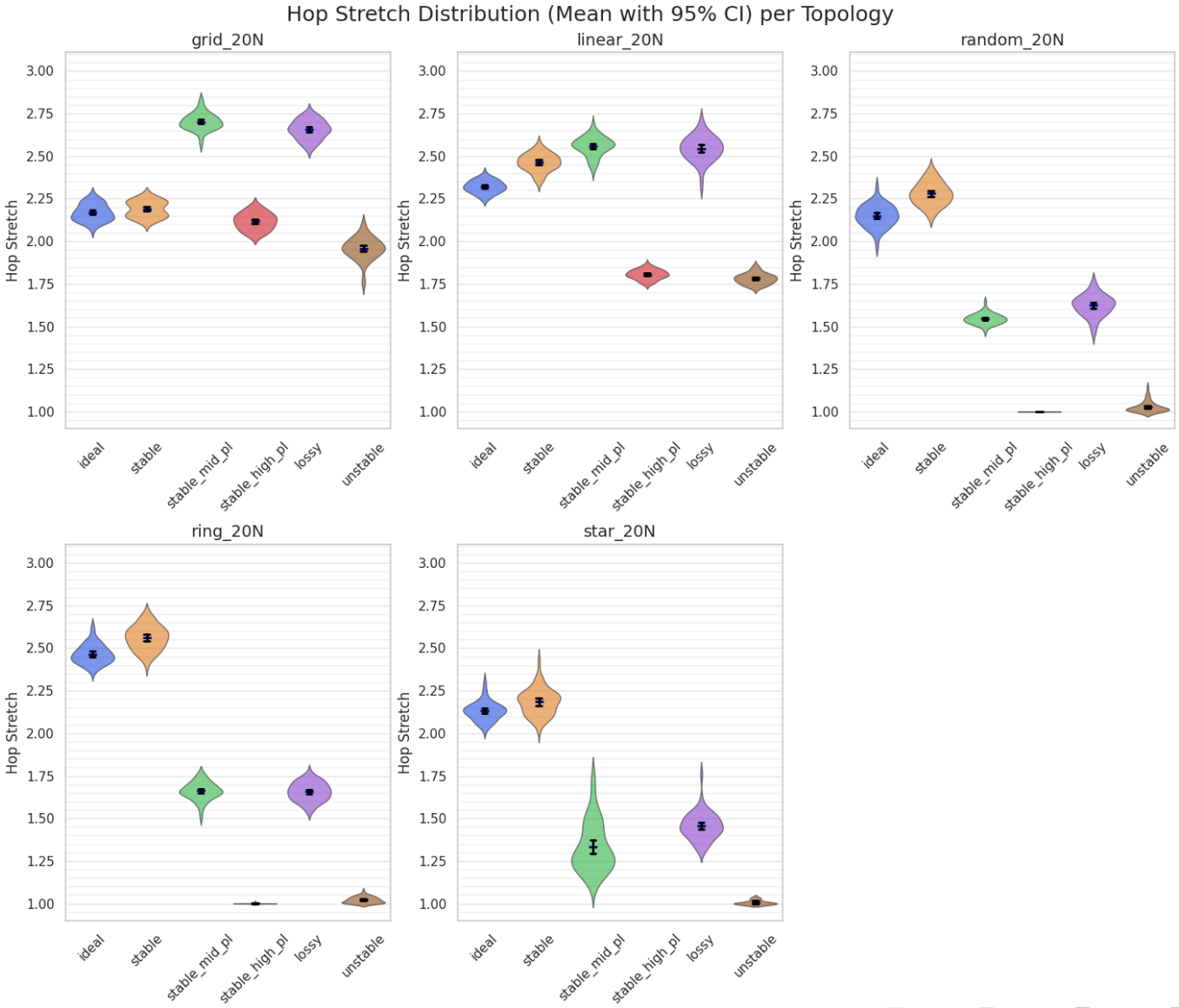
Results: Packet Delivery Ratio (PDR)



Results: End-to-End Latency (E2E)



Results: Hop Stretch (HS)



Conclusions

- Grid and Linear topologies show strongest resilience
- Star topology highly sensitive to path loss
- Channel instability reduces connectivity